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Parsing for Prosody: What a Text-to-Speech System Needs from Syntax

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Abstract

We describe an experimental text-to-speech system that uses a syntactic parser and prosody rules to determine prosodic phrasing for synthesized speech. Our results indicate that many aspects of sentence analysis that are required for other parsing applications, e.g. machine translation and question answering, become unnecessary in parsing for text-to-speech. It is possible to generate natural-sounding prosodic phrasing by relying on information about syntactic category type, partial constituency, and length; information about clausal and verb phrase constituency, predicate-argument relations, and prepositional phrase attachment can be bypassed.

Introduction

Most text-to-speech systems include two components: pronunciation rules and a speech synthesizer. Pronunciation rules convert input text into a phonetic transcription, possibly supplementing the process with a dictionary that provides information about the part of speech, stress pattern, and phonetic makeup of particular words. The speech synthesizer then converts the phonetic transcription into a series of speech parameters that are subsequently processed to produce digitized speech. These systems tend to perform well on word pronunciation but fall short when it comes to providing good prosody for complete sentences. One reason for this shortcoming is that, while sentence prosody has been shown to be influenced by various aspects of syntactic structure, the text-to-speech field has lacked a robust working system to test the possible relations between syntax and sentence prosody.

In this paper, we describe an experimental text-to-speech system that uses a syntactic parser and prosody rules to specify the location and relative strength of prosodic phrase boundaries for English input. This information is passed to the Bell Labs text-to-speech programs [15], which assign a pitch contour, segment duration, and pausing specifications over the domain specified by our system. The goal of our system has been to produce synthetic speech that has natural sounding prosodic phrasing.

At the outset of the project we had available to us a moderate-coverage deterministic parser, Fidditch [9]. The central question that we posed with respect to this parser was: How much parsing information is necessary to determine the location of prosodic phrase boundaries. Assuming that the prosodic phrasing of speech conveys certain syntactic properties of the sentence to the listener, our intention was to learn whether a moderate coverage parser would provide the syntactic information necessary for a text-to-

speech implementation, or whether there would be some amount of over- or under-specification in fitting the syntax and the speech together.

Our results indicate that while prosody makes use of some syntactic information, many of the characteristics of the sentence that are described by parsing systems designed for information retrieval, machine translation, or text generation are unnecessary in parsing for text-to-speech.¹ Section 2 outlines the problem in interfacing syntax and prosody, section 3 discusses the methods we used to evaluate our system and briefly describes the components of the system, with emphasis on its parser, and section 4 describes those aspects of syntax that we found necessary for prosodic phrasing and those that are discarded by the phrasing.

The Syntax/Prosody Relation

Although some connections between syntax and prosody are well-known, e.g., the influence of part of speech on stress in differentiating the noun *'progress* from the verb *pro'gress*, very little practical knowledge is available on which aspects of syntax might be connected to prosodic phrasing. Linguistic work on the relation between syntax and phrasing has noted that prosodic pauses do not consistently align with syntactic constituent edges. For instance, Chomsky and Halle [3] point out that the syntactic bracketing of their example (1a) does not correspond to the prosodic phrasing (1b).

1. a. This is [_{NP} the cat that caught [_{NP} the rat that stole [_{NP} the cheese]]]
b. This is the cat -- that caught the rat -- that stole the cheese

Chomsky and Halle dismiss such misalignments as "...a performance factor, related to the difficulty of producing right branching structures such as [1]." However, research beginning with Martin [13] and including Grosjean, Grosjean, and Lane [8], Dommergues and Grosjean [4], Umeda [18], Gee and Grosjean [7], and Nespor and Vogel [14] has indicated consistent patterns of misalignment that invite experimental testing of a large corpus.²

1. On the other hand, extrasyntactic phenomena such as parentheticals, parallelism, listing, coreference, and contrast are crucial both to the determination of prosodic phrasing and to the assignment of the appropriate pitch contour. We limited ourselves initially to those characteristics of text-to-speech that stem from the syntactic and phonological properties of the sentence, in order to make appropriate use of the tool we had at hand, namely the Fidditch parser, and to avoid, in our initial implementation, problems that are famous for their intractability.
2. Umeda [18] provides a rather thorough delineation of the correlation between syntactic constituency and prosodic phrasing in a study involving 3549 words of text, 5 speakers, and 9 listeners. She also provides reasons, such as variations in information density, for the lack of a correlation when such

Our goal was to develop a theory of syntax/prosody relations that we could test in an experimental text-to-speech system. To our knowledge, no prior studies have coupled a moderate-to-large coverage parser with synthesized speech for the purpose of investigating sentence prosody.³ We also hoped to further the state of the art in speech technology. For many unpracticed listeners, texts of only moderate difficulty can often be incomprehensible when spoken by a synthesizer. Several authors (e.g. [1], [5], [10]) have suggested that prosodic differences between synthetic and natural speech are the primary unaddressed factor leading to difficulties in the comprehension of fluent synthetic speech.

Overview of the System

Our system comprises three components: (i) the parser, which builds syntactic structure, (ii) a set of prosody rules that generate a prosodic structure using the syntactic structure, and (iii) the AT&T Bell Labs text-to-speech synthesizer [15]. The parser and speech programs are independent components. The prosody rules act as a filter between them, converting the syntactic information generated by the parser into annotations of the text input, which is then supplied to the text-to-speech system in the form of escape mechanisms that specify changes in pitch contour, segment duration, and boundary tone.

Our approach follows the work of [7] in its reliance on the notions of constituent length and balancing in the syntax/prosody interface. We have also drawn from the prosodic theory outlined in [17].

Testing

Our current system is a revision of a prototype tested against a corpus of 39 sentences which included text from a repair manual for telephone switching systems, an introductory description of the *Prose 2000* text-to-speech system, sentences from [18], and sentences that we composed in order to test a good range of English syntactic constructions. Although our overall results were very good, we detected patterns of failure with two types of syntactic constructions: (1) embedded sentences were set off prosodically, as in *They believe -- California sales are still off 75 percent*, even when informant testing indicated that this was inappropriate; and (2) adjuncts were set off unnaturally, as in *The speaker pronounced the names of the characters -- on the left*.

Because the failures we encountered were not random, we decided to transcribe spoken texts to discover what should be happening with the problematic constructions. Both of us transcribed two professionally produced tapes, a rendition of Sir Arthur Conan Doyle's *The Speckled Band* and *Everest! A Sound History 1922-1983*, which contains taped interviews as well as narration. We marked the transcriptions independently for pausing and pitch contour information, compared the markings, and discarded the discrepancies. Our evidence as to what the prosodic phrasing needs from the syntax is taken from these annotated transcriptions.

Parsing

The parser identifies syntactic branching patterns and node labels for each input sentence. Our parser is a version of Fidditch [9], a moderate coverage parser based on the deterministic model described in [11]. To build syntactic structure, the parser uses a

occurs. However, she does not match up the reasons with the specific syntactic situations that she discusses, leaving many open questions as to how syntax interfaces with prosody.

3. Umeda and Teranishi [19] describe a "simple syntactic analyzer" designed for test-to-speech but do not discuss what the program reveals about specific syntax-prosody connections.

grammar that requires the representations produced by lexical and syntactic rules to be consistent with semantics. The grammar is modeled on the government and binding framework of generative grammar, as adapted to the requirements of the parser ([2], [6]).

Input to the parser is a sentence or phrase. Its output is a surface structure tree that identifies syntactic constituents, parts of speech, and null terminals like the gap associated with *Who* in *Who did you speak to ___*? As it builds the tree, the parser distinguishes grammatical roles such as subject, complement, head, and adjunct by representing a constituent as either attached or unattached to higher nodes. Each sentence is analyzed as a "core" *S* consisting of subject (*NP*), modality (*AUX*), and predicate (*VP*). Modifiers and adjuncts, which are optional constituents, are excluded from the core. Consequently, only elements that belong in the core are attached to the *S* node; adjuncts and modifiers have no assigned structural position within *S* and so are represented as unattached "orphan" nodes. Similarly, within a phrase, only the head, its pre-head modifiers, and complements of the head are attached; relative clauses, comparatives, adverb phrases, and any other "non-argument" constituents, whose attachment is partially semantically and pragmatically determined [11], go unattached.

Figure 1 shows a typical syntax tree. Terminal elements of the tree are words. Other possible terminals are null elements inserted by the parser, which include null tense elements under *AUX* and two types of null *NP*, * and *DELTA*. The null *NPs*, which stand for understood subjects, are available for semantic processing but are deleted by the prosody rules. The encircled node labels of Figure 1 indicate the constituent type of the structure headed by that node. An encircled empty node is the parser's indication that that node's attachment is not syntactically determinable. This is the situation with the node dominating the adverb *only*. Hence the parse tree makes no claim about which constituent *only* modifies.

Prosody Rules

Our prosody rules operate in three stages. The first stage identifies possible prosodic phrases. The second stage decides which phrases will be realized in speech by assigning each phrase boundary a relative strength (or perceptibility). The last stage specifies an acoustic value for each boundary strength. In (2), for example, prosodic phrase boundaries occur after *off*, and *character*,

2. When this switch is off--the name of the character--is not pronounced.

When relative strengths are assigned, the boundary following *off* receives the highest strength value, which involves pausing, final lengthening, and pitch perturbation.

Adjunction and Phonological Phrasing Rules. Adjunction applies first to the terminal nodes in the syntax tree, combining certain orthographically distinct words into phonological constituents that have no internal word boundaries. They merge a word with a left or right neighbor, according to (i) the category of the word, and (ii) its position in the tree. In general, the adjoinable words are function words: complementizers, articles, conjunctions, auxiliary verbs, prepositions and pronouns [17]. Whether a word adjoins to the left or the right depends on its grammatical relations with surrounding material. For example, a preposition always adjoins rightward when it is the head of a *PP*; thus *on* merges with *printed* in *on printed wiring board*. However, when the preposition is a sister of the verb in the syntax tree, it adjoins leftward, as does *on* in *Such communication relies on voice response*. Constituents created by adjunction are treated as

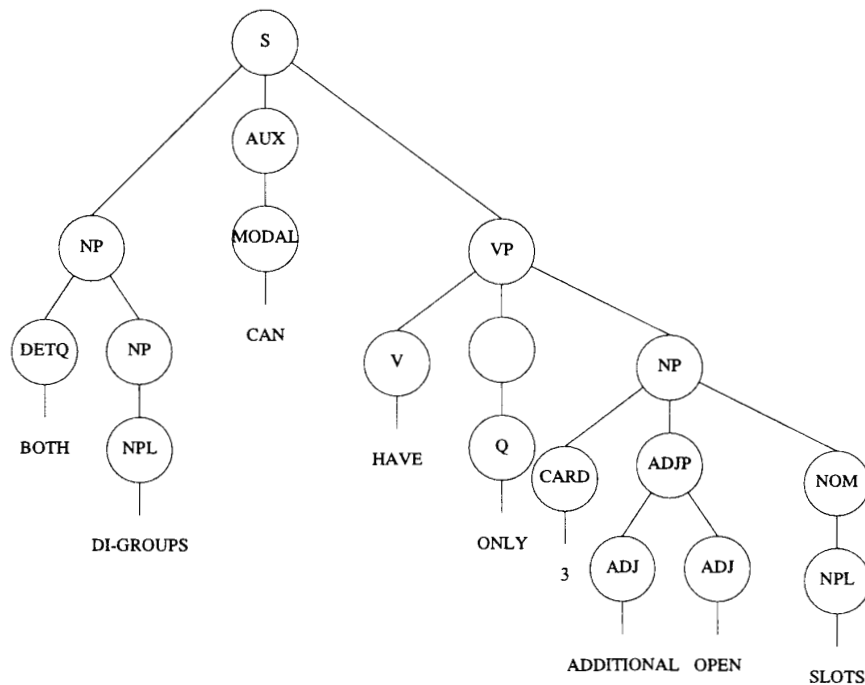


Figure 1: Parse Tree

Both di-groups can have only 3 additional open slots.

phonological words by subsequent rules. Thus *on printed* and *relies on* each count as a single word for later balancing and length rules, and the synthesizer will avoid inserting prosodic word boundary cues into such constituents.

Phonological phrasing rules build the results of adjunction into larger phonological phrases (equivalent to the *phi* phrases of [7]). Phonological phrases, in our system, are groups of words that cohere strongly in speech and thus should not be separated by phrase boundaries. They consist of a syntactic head plus the material that intervenes between it and a preceding head.

Each syntactic head in Figure 1 signals the right edge of a phonological phrase and, in so doing, identifies the site of a possible prosodic phrase boundary. The sample sentence of Figure 1 contains 3 phonological phrases, based on the occurrence of syntactic heads: [*Both di-groups*], [*can have*], and [*only 3 additional open slots*]

Phrase Realization Rules. The prosody phrasing rules of stage 2 decide which phonological boundaries will delimit prosodic phrases in speech. Our current system employs three prosody phrasing rules:

(i) **Verb balancing** groups a verb to the left or to the right in order to produce two prosodic phrases of roughly equal length. The rule, which is an adaptation of that given in [7], is:

in $X \ V \ Y$
 if $\text{length}(X) + \text{length}(V) < \text{length}(Y)$,
 then V groups with X
 otherwise V groups with Y
 where X and Y are phonological phrases

For example, in (3) the verb groups leftward because *both di-groups* on the left, is shorter than *only three additional open slots*; in (4) the verb groups rightward because *two di-groups* on the right is shorter than *each shelf of the 48-channel module*.

3. both di-groups can have -- only three additional open slots
4. Each shelf of the 48-channel module -- has two di-groups

(ii) **Verb adjacency** applies after the balancing rule, bundling the constituent left over from the balancing with the prosodic phrase containing the verb. Applied to (4), it bundles *each shelf of the 48-channel module* with *has two di-groups* so that the constituents within this phrase cohere more closely with each other than with the other constituents in the sentence.

(iii) **Length rules** apply to phrases that are on the periphery of the cluster formed by (ii) to prosodically set off longer phrases. Our current length rules are limited to PP constituents, and assign greater prosodic prominence (i.e. a higher boundary strength) to constituents whose phonological word length is greater than 1,

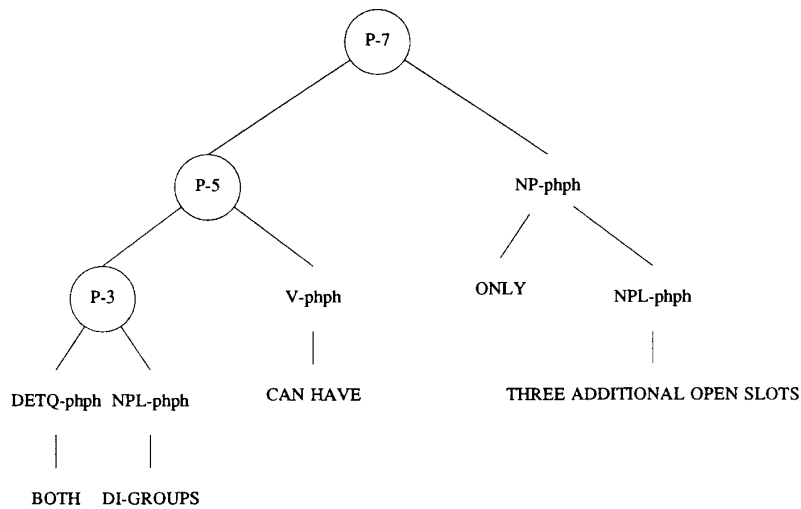


Figure 2: Prosody Tree
(phph = phonological phrase, P = prosodic phrase)

Both di-groups can have only 3 additional open slots.

where a 'phonological word' is made up of a lexical word and any adjoined words. Thus, *in the village* in (5) does not form a separate prosodic phrase, but *at the village inn* in (6) does. (The double vertical line indicates a more perceptible boundary -- in our data a pause -- than does the single vertical line.)

5. He's ended || by becoming | dreadfully feared in the village.
6. We had no difficulty || in engaging rooms | at the village inn.

Finally, any remaining constituents are simply bundled from left to right, into a binary tree [7].

The prosody tree in Figure 2 shows the results of adjunction, phonological phrasing, and two prosody rules: verb balancing and verb adjacency. The numbered nodes in this tree mark prosodic phrase boundaries. The indices, which are based on phonological word counts, refer to the relative strength of a phrase boundary.

Prosody Conversion Rules. The rules that determine acoustic values map the boundary strength indices determined by the phrasing rules onto three phonological mechanisms. Boundary indices in the low range, are realized as a phrase accent. Mid-range indices are realized as modulations in pitch. High indices are realized as changes in both pitch contour and duration. Pitch contour determinations are based on the model of [16]. Thus the hierarchical organization of a structure such as that in Figure 2 can be reflected directly in the synthesized speech.

How Much Syntactic Information does Prosodic Phrasing Need?

Our parser provides the prosody rules of our system with the following necessary information: syntactic category status, location of the head of a syntactic phrase, noun phrase and prepositional phrase constituency, and the location of gaps that are associated

with *wh-* items. This information makes a parser a necessary part of a speech system that includes natural sounding prosodic phrasing.

The parser overgenerates with respect to prosody because it provides the following unnecessary information: sentence and verb phrase constituency,⁴ predicate-argument relations (which are represented in the syntax tree by dominance relations) and the location of non-*wh-* gaps. In addition, although the parser does not attach adjunct phrases, parsers designed for information retrieval do; such attachment is also ignored by prosodic phrasing.

The Parsing Information Used by Text-to-Speech. Our rules require syntactic lexical category information to identify function words, which undergo adjunction, and verbs, which initiate the verb balancing rule. In addition, syntactic head information is needed for adjunction and for the identification of phonological phrases.

The phrase realization rules require the identification of noun phrases and prepositional phrases. Evidence of the need for noun phrase recognition comes from comparing our data with the balancing required around the verb. Prepositional phrase recognition is necessary in order to identify where an introductory PP ends and the subject of the core sentence begins.

Because of examples like (7), we are assuming that all the material dominated by a noun phrase is counted in balancing the constituents on either side of the verb.⁵ In (7) the phrase *curling up*

4. Full adjective and adverb phrases occur in our data as predicate phrases, e.g., *It was [ready to bite]*. The verb involved in these phrases is usually some form of *be*, whose function as a full verb is unclear. We therefore leave the prosodic status of adjectival and adverbial phrases as an open question.

5. Unfortunately testing the verb rule against our transcribed text yields only a small amount of data since examples in which either the subject is long and the object short, or vice-versa, are unusual.

from the chimneys is 2 phonological words -- shorter than the 4 phonological words of *that part of the west wing*.

7. blue smoke | curling up from the chimneys || showed that part of the west wing || in which the family resided.

Nevertheless, the verb groups with the latter phrase. This grouping is accounted for if the verb rule looks at the whole noun phrase, *blue smoke curling up from the chimneys* when balancing the constituents on either side of the verb. The example (8) makes the same point except that here the verb *bite* coheres with the phrase to its left.

8. it was ready to bite || the first person it saw

Again, if the verb rule looks at the whole noun phrase *the first person it saw* the grouping is accounted for.⁶

Prepositional phrases must also be recognized by the parser. PP recognition is needed for sentences in which a subordinate PP precedes a core sentence, as in (9):

9. Before John told us he knew, the answer was supplied.

The prosodic phrasing for (9) groups *knew* with the PP *Before John told us he knew* and the noun phrase *the answer* is grouped with the verb phrase. The parser must therefore supply the information to the prosodic phrasing that *knew* is part of the initial PP, but that *the answer* is not.⁷

The parser must also inform the prosody of the location of gaps that are associated with the *wh-* items *who*, *what*, *where*, etc. It appears from the small amount of data we have (six examples) that such gaps participate in the word count in prosodic rules involving length; in our system these are the verb rule and the prepositional phrase rule. This would explain the prosodic difference between (10) and (11), where the gap in (11) prevents the verb *do* from grouping with *yourself*.

10. What are you going to give yourself _ ?
11. What are you going to do _ yourself ?

The Parsing Information Not Used by Text-to-Speech. Our parser currently provides more information than needed for prosody, as will any parser that provides a detailed syntactic tree. In fact, any parser whose syntax is designed with some level of semantic processing in view will overgenerate with respect to prosody since not all of the syntactic information necessary for semantic analysis is necessary for prosodic analysis. The information not required by prosody includes the identification of VP and S constituency, the recognition of predicate-argument relations, scope disambiguation and, in some parsers, the recognition of null elements other than the trace of *wh-* as placeholders for arguments.

Examples like (12) show that the VP node is ignored by prosody:

6. There are data in which all the material under the NP is not considered by the verb rule, as (i) and (ii) illustrate:
 - (i) Holmes || examined the outsides of the three bedroom windows
 - (ii) From outside || came the occasional cry of a bird of the night
 These exceptions to the verb rule's NP count appear to be due to extraneous yet predictable factors. In cases like (i), we assume that the subject is set off from the verb because it is a proper name, and in (ii), the verb's grouping with the longer phrase appears to be due to the highlighting of the prepositional phrase *from outside* caused by the inversion of *The occasional cry of a bird of the night came from outside*.
7. Instead of recognizing where the initial PP ends, an alternate parser might recognize where the NP subject begins. At present we have no motivation within our system for choosing this approach over the phrase structure approach.

12. a. Hunt despised || this patriotic fervor.
- b. We had to do || all the work of the house.

In (12a), for example, the VP constituent is broken up, with the verb *despised* going to the left of the prosodic break and the verb complement *this patriotic fervor* going to the right. The verb thus groups with the shorter of its two neighbors, illustrating that the notion of balancing constituents in terms of length, rather than VP constituency, is at work in deciding prosodic phrasing around the verb.

The S node is similarly ignored in prosodic phrasing. Examples like (13), with a dependent S, *that an avalanche was beginning to fall*, appear to acknowledge the S node:

13. I realized at once || that an avalanche was beginning to fall.

However, according to our analysis, the perceptible break before the dependent sentence in (13) results from the balancing of constituents around the verb, and not from the presence of the S node. In (13), the pronoun *I* adjoins to *realized* and the phrase *at once* groups with *realized* by the verb adjacency rule. The material under the dependent S forms a prosodic constituent via verb balancing and verb adjacency, in which *to fall* groups with *was beginning* by verb balancing, *that* and *an* adjoin to *avalanche* and *that an avalanche* groups with *was beginning to fall* via verb adjacency. This prosodic constituent is attached to the prosodic tree by left-to-right bundling. Bundling, as the last stage in the mapping from the syntactic tree to the prosodic tree, is associated with the highest boundary strength. Therefore, our analysis claims that the prosodic break before the dependent S in (13) results from verb balancing and verb adjacency, rather than from the prosodic recognition of the dependent S.

Strong support for our claim that the S node is not recognized in prosodic phrasing comes from examples like (14)-(15) where there is no correspondence between the syntactic S (in italics) and the prosodic phrasing:

14. Even my fiancé | believes *it's only my imagination*.
15. He knew *the first major confrontation* || *was the ice fall*.

Here again the phrasing is the effect of verb balancing and verb adjacency: in (14) since the phrase *Even my fiancé* is not short enough, the verb balancing rule groups *believes* with the phrase to its right, creating no prosodic break between the verb and the embedded S; in (15) *he* adjoins to the verb *knew* and the phrase *the first major confrontation* groups with *He knew* via verb adjacency. The resulting phrase joins finally with *was the ice fall* via verb adjacency. Since this is the last step in the phrasing, this juncture receives the highest boundary strength, accounting for the break in (15). The fact that the break occurs in the middle of the dependent S rather than at its beginning illustrates that the dependent S is ignored by prosodic phrasing.

Prosodic phrasing also ignores the predicate-argument relations. The grouping of the embedded subject in (15) with the verb of the higher sentence rather than with the S that dominates it illustrates that the subject-of relation does not have to be recognized by prosodic phrasing. Similarly, if the complement-of and adjunct-of relations were recognized by prosodic phrasing, we would expect complements to group with the head verb and adjuncts to be set off. However, the facts are the reverse when the complement is longer than the phrase to the left of the verb, as in (16)-(17), or when the adjunct is short, i.e. a single phonological word, as in (18)-(19):

16. Hunt despised || this patriotic fervor.
17. she died || of sheer terror and nervous shock
18. Seven of our porters || were killed in the fall.
19. he must be asleep by now.

Null elements have no effect on prosodic phrasing, except for the gap associated with *wh*-question words, discussed in section 4.1. If the null DELTA in (20), which is linked to the subject *I*, counted prosodically, we would expect a break after the verb *seem*.

20. I seem DELTA to see dimly || what you're driving at.

The fact that none occurs in such cases indicates that these elements are not prosodically significant.

Finally, prosodic phrasing can gloss over scope ambiguities that present well-known difficulties to parsers designed for semantic analysis. The following sentence from our data is typical of such ambiguity:

21. Holmes || had brought a long thin cane with him || which he laid | on the bed | beside him.

In (21), either the bed or the cane is beside Holmes. A parser that chooses to capture the ambiguity attaches the phrase *beside him* under the NP dominating *the bed* for the former reading and higher in the tree for the latter. Prosodically, the single vertical line after *bed* in (21) indicates a pitch perturbation but no pausing and, to our ears, no prosodic clue is given as to which reading is intended.

The absence of a prosodic disambiguation clue in (21) is consistent with the claim of [12] that the attachment of adjuncts in syntax is *appropriately vague*. This vagueness creates the effect that while such sentences are believed to be understood, the "vagueness may not be resolved until a hearer's attention is called to the unresolved decision." In the same view, [14] provides experimental evidence that prosodic phrasing does not tap into prepositional phrase attachment. Thirty-six Italian-speaking adults, when asked to give the meaning of sentences of the type (22), were unable to guess whether the speaker intended the meaning in (a) or that in (b), even though the speaker modelled the intended reading on a non-ambiguous sentence with the intention of either (a) or (b).⁸

22. Marco ha guardata la ragazza col canocchiale.
(Marco looked at the girl with the binoculars)
a. The girl is holding the binoculars.
b. Marco is holding the binoculars.

Our parser, which is based on the Marcus model, represents the internal structure of an adjunct but does not attach it to the syntactic tree, the assumption being that the relation of an adjunct to the sentence is determined by semantic and pragmatic factors, not by syntactic attachment. We assume then that adjunct attachment differs from the other factors that are ignored by prosody in that attachment ambiguities are not part of natural language syntax, despite the fact that many parsers choose to capture the ambiguity with a syntactic mechanism.

8. The subjects chose the intended meaning of prepositional phrase attachment ambiguities like (22) only 59.1% of the time. This compares to 90.2% of the time for sentences like (i) which involve ambiguities other than PP attachment:

(i) Li suonano insieme.
a. There they are playing together.
b. They are playing them together.

Conclusion

We have described an experimental system that uses a natural language parser and three levels of prosody rules to generate prosodic phrases for synthetic speech. In developing rules for the system, we have taken the approach that phrasing depends on a process of balancing and length assessment that may or may not be consistent with syntactic structure. The main contribution of our system to the study of prosodic phrasing is that it allows us to characterize, precisely and for an unrestricted range of English, where syntactic and prosodic structures converge and where they differ.

Specifically, we found that prosodic phrasing requires information from the syntax tree that sits rather low down on the tree. This includes part-of-speech information, syntactic head information, information on the material under NPs and PPs, and information about the gaps associated with *wh*-question words. The need for this information on the part of prosodic phrasing makes some amount of parsing a necessary part of a speech system that includes prosody.

In contrast, prosodic phrasing generally ignores information from the higher levels of the syntax tree. This includes information on S and VP constituency, predicate-argument relations, and adjunct prepositional phrase attachment. Information on empty nodes other than *wh*-gaps are also ignored. This is information that has been included in the syntax tree to expedite semantic processing.

The testing of this prosodic system has made clear to us that mechanisms built into a parser to enhance semantic processing often only complicate the mapping from syntax to prosodic phrasing. Our current goal is to reexamine the types of alignment and misalignment between syntax and prosodic phrasing in an attempt to find a pattern that will be the basis for a more efficient text-to-speech parser that avoids the syntactic overgeneration of the current parser.

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